

I use two data sources:

County Population Totals and Components of Change: 2020- 2025

[Excel file]([https://www2.census.gov/programs-surveys/popest/tables/2020- 2025/counties/totals/co-est2025-pop.xlsx](https://www2.census.gov/programs-surveys/popest/tables/2020-2025/counties/totals/co-est2025-pop.xlsx))

Zillow Housing Data

[csv file](https://files.zillowstatic.com/research/public_csvs/zhvi/County_zhvi_uc_sfrcondo_tier_0.33_0.67_sm_sa_month.csv?t=1775837851)

The code below imports county-level data from Zillow (named "housing") and the Census Bureau (named "pop"). For housing, we use the data from December 2020. In the dataset, housing prices are labeled as "hp" and population data is labeled as "pop".

```
In [1]: suppressWarnings(suppressMessages({
library(dplyr); library(ggplot2); library(readxl); library(ggrepel); options(warn = -1)
options(repr.plot.width = 15, repr.plot.height = 9)
zillow_data = 'https://files.zillowstatic.com/research/public_csvs/zhvi/County_zhvi_uc_sfrcondo_tier_0.33_0.67_sm_sa_
pop_data = 'https://www2.census.gov/programs-surveys/popest/tables/2020-2025/counties/totals/co-est2025-pop.xlsx'
housing = read.csv(zillow_data) %>% mutate(State = state.name[match(State, state.abb)],
name = paste0(RegionName, ', ', State)) %>%
select(name, contains("12.31") & contains("202")) %>%
rename_with(~ paste0("hp", 2020:2025), 2:7) %>% na.omit()
ct_crosswalk = data.frame( # Connecticut County names
planning_region = c("Capitol Planning Region, Connecticut", "Greater Bridgeport Planning Region, Connecticut",
"Lower Connecticut River Valley Planning Region, Connecticut", "Naugatuck Valley Planning Region, Connecticut",
"Northeastern Connecticut Planning Region, Connecticut", "Northwest Hills Planning Region, Connecticut",
"South Central Connecticut Planning Region, Connecticut", "Southeastern Connecticut Planning Region, Connecticut",
"Western Connecticut Planning Region, Connecticut"),
legacy_county = c(
"Hartford County, Connecticut", "Fairfield County, Connecticut", "Middlesex County, Connecticut",
"New Haven County, Connecticut", "Windham County, Connecticut", "Litchfield County, Connecticut",
"New Haven County, Connecticut", "New London County, Connecticut", "Fairfield County, Connecticut"))
housing = housing %>% left_join(ct_crosswalk, by = c("name" = "legacy_county")) %>%
mutate(name = ifelse(!is.na(planning_region), planning_region, name)) %>% select(-planning_region);
temp_file = tempfile(fileext = ".xlsx")
download.file(pop_data, destfile = temp_file, mode = "wb")
pop = read_xlsx(temp_file, skip = 3)[-1, -2] %>%
rename_with(~ paste0("pop", 2020:2025), 2:7) %>% rename(name = 1) %>%
mutate(name = gsub('^\\.\\.', '', name)) %>% na.omit()))
```

Q1. Which counties experienced the largest increases or decreases in housing prices and population? This code calculates the change in housing prices and population between 2020 and 2025. It then identifies and displays the top 5 counties with the highest and lowest changes for both variables.

```
In [2]: housing$hp_change = housing$hp2025 - housing$hp2020
pop$pop_change = pop$pop2025 - pop$pop2020
housing %>% arrange(hp_change) %>% head(5)
housing %>% arrange(hp_change) %>% tail(5)
pop %>% arrange(pop_change) %>% head(5)
pop %>% arrange(pop_change) %>% tail(5)
```

A data.frame: 5 × 8

	name	hp2020	hp2021	hp2022	hp2023	hp2024	hp2025	hp_change
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	Chariton County, Missouri	257651.6	260413.5	169930.1	143494.1	131695.16	114444.67	-143206.92
2	New York County, New York	1311302.5	1348813.9	1319081.3	1149015.6	1170737.91	1178821.26	-132481.27
3	Claiborne Parish, Louisiana	221569.8	159580.2	144500.3	122085.3	91590.78	89377.97	-132191.86
4	Wells County, North Dakota	227897.3	201497.6	203140.7	167374.1	160837.95	130024.11	-97873.15
5	Trigg County, Kentucky	238998.4	213961.1	213682.8	178911.9	164833.74	148865.73	-90132.70

A data.frame: 5 × 8

	name	hp2020	hp2021	hp2022	hp2023	hp2024	hp2025	hp_change
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
3006	Dukes County, Massachusetts	978456.7	1219809	1401661	1454702	1486242	1497809	519352.7
3007	San Miguel County, Colorado	937739.2	1314023	1518842	1422515	1463383	1526008	588269.1
3008	Teton County, Wyoming	1240005.3	1739402	1948342	2187646	2149942	2124505	884499.8
3009	Nantucket County, Massachusetts	1981341.1	2523684	2753583	2822330	2913281	2963899	982558.1
3010	Pitkin County, Colorado	1469758.2	1993348	2236881	2244061	2463566	2566677	1096918.3

A tibble: 5 × 8

name	pop2020	pop2021	pop2022	pop2023	pop2024	pop2025	pop_change
<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
Los Angeles County, California	9996510	9810478	9748524	9732568	9748868	9694934	-301576
Cook County, Illinois	5263651	5186249	5127866	5142853	5188791	5194625	-69026
Kings County, New York	2722259	2634576	2600751	2611893	2658657	2653963	-68296
Bronx County, New York	1461233	1420980	1384053	1380478	1406052	1406332	-54901
San Francisco County, California	875139	816169	813266	816926	820945	826079	-49060

A tibble: 5 × 8

name	pop2020	pop2021	pop2022	pop2023	pop2024	pop2025	pop_change
<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
Denton County, Texas	914517	944283	980835	1013965	1046226	1069346	154829
Montgomery County, Texas	625369	650934	681285	715961	751183	781194	155825
Collin County, Texas	1075512	1114667	1163646	1207386	1254213	1297179	221667
Maricopa County, Arizona	4445025	4500059	4559561	4601077	4654147	4689558	244533
Harris County, Texas	4734060	4736539	4809691	4896507	4996331	5045026	310966

```
In [3]: # Challenge:
# Since raw numerical differences are not useful, replace with code that calculates percentage changes instead.
housing$hp_change = ((housing$hp2025 - housing$hp2020) / housing$hp2020) * 100
pop$pop_change = ((pop$pop2025 - pop$pop2020) / pop$pop2020) * 100
housing %>% arrange(hp_change) %>% head(10)
housing %>% arrange(hp_change) %>% tail(10)
pop %>% arrange(pop_change) %>% head(10)
pop %>% arrange(pop_change) %>% tail(10)
```

A data.frame: 10 × 8

	name	hp2020	hp2021	hp2022	hp2023	hp2024	hp2025	hp_change
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	Claiborne Parish, Louisiana	221569.83	159580.20	144500.27	122085.26	91590.78	89377.97	-59.66149
2	Chariton County, Missouri	257651.59	260413.52	169930.09	143494.07	131695.16	114444.67	-55.58162
3	Washington County, Mississippi	127077.55	131800.70	103480.74	94336.85	85719.84	61246.84	-51.80357
4	Wells County, North Dakota	227897.26	201497.63	203140.68	167374.14	160837.95	130024.11	-42.94617
5	Scott County, Illinois	143379.18	130258.28	108396.68	99604.80	99781.89	85767.32	-40.18147
6	Lamoure County, North Dakota	224148.54	199364.49	195775.90	161997.14	163730.04	138553.42	-38.18678
7	Trigg County, Kentucky	238998.44	213961.06	213682.77	178911.88	164833.74	148865.73	-37.71267
8	Floyd County, Texas	131715.57	122801.93	116079.59	97377.69	94140.87	82890.11	-37.06886
9	McDowell County, West Virginia	64542.03	70394.42	59904.16	52872.66	55775.29	42354.39	-34.37705
10	Reeves County, Texas	259781.89	287215.34	257643.56	186917.02	175573.07	170735.94	-34.27719

A data.frame: 10 × 8

	name	hp2020	hp2021	hp2022	hp2023	hp2024	hp2025	hp_change
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
3001	Martinsville City, Virginia	67545.08	78853.23	96181.27	95626.22	111911.4	116052.5	71.81486
3002	Sweet Grass County, Montana	271395.63	342681.60	385426.95	395038.02	435220.2	468962.6	72.79665
3003	Rich County, Utah	335301.90	468624.13	523285.46	530303.18	543710.9	580020.5	72.98455
3004	Pitkin County, Colorado	1469758.16	1993348.37	2236880.51	2244060.82	2463566.0	2566676.5	74.63257
3005	Blaine County, Idaho	599172.97	814481.39	887024.73	888702.93	921895.0	1049191.0	75.10653
3006	Deer Lodge County, Montana	154750.75	205762.30	243342.17	258187.60	278180.7	281125.2	81.66323
3007	Watonwan County, Minnesota	91078.95	117425.08	132065.40	146654.31	160065.9	169454.0	86.05172
3008	Powder River County, Montana	152357.78	194580.17	232713.54	235875.49	258607.7	285640.6	87.48017
3009	Petersburg City, Virginia	99598.64	122669.50	198553.57	188273.63	196915.6	191325.3	92.09633
3010	Garfield County, Montana	149446.14	193460.54	219920.42	224882.68	275008.5	303193.9	102.87835

A tibble: 10 × 8

name	pop2020	pop2021	pop2022	pop2023	pop2024	pop2025	pop_change
<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
King County, Texas	267	253	231	212	212	192	-28.08989
Garza County, Texas	5806	6257	4502	4544	4567	4510	-22.32174
Sharkey County, Mississippi	3773	3629	3484	3375	3199	3097	-17.91678
Reeves County, Texas	14769	13015	11771	11937	12125	12138	-17.81434
Loving County, Texas	63	57	46	46	47	52	-17.46032
Cameron Parish, Louisiana	5621	5094	4932	4821	4773	4718	-16.06476
DeKalb County, Missouri	11605	11700	11299	9893	9882	9795	-15.59673
Tallahatchie County, Mississippi	12652	11945	11730	11476	10961	10877	-14.02940
Yazoo County, Mississippi	26671	25181	24699	24584	23166	22947	-13.96273
Jeff Davis County, Texas	1987	1963	1909	1852	1775	1711	-13.89029

A tibble: 10 × 8

	name	pop2020	pop2021	pop2022	pop2023	pop2024	pop2025	pop_change
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
	Pinal County, Arizona	430231	449128	465550	486554	515726	539380	25.36986
	Brunswick County, North Carolina	138135	144719	152519	159425	166889	174702	26.47193
	Comal County, Texas	163608	174878	184600	193849	201486	209166	27.84583
	Rockwall County, Texas	109199	116645	123439	131418	136880	140738	28.88213
	Jackson County, Georgia	76765	80421	84157	88891	94302	99265	29.31023
	Dawson County, Georgia	27062	28527	30237	31833	33945	35365	30.68140
	Liberty County, Texas	92471	97404	102615	109248	116291	121364	31.24547
	Jasper County, South Carolina	29158	30441	32068	33983	36338	38533	32.15241
	Long County, Georgia	16256	17071	18363	19608	20493	21557	32.60950
	Kaufman County, Texas	147165	158316	172773	186658	198010	209235	42.17715

Q2. Which counties experienced the largest percentage increases or decreases in housing prices and population?

```
In [4]: data = housing %>% left_join(pop, by = 'name') %>% na.omit()
head(data)

model = lm(hp_change ~ pop_change, data = data)
summary(model)
```

A data.frame: 6 × 15

	name	hp2020	hp2021	hp2022	hp2023	hp2024	hp2025	hp_change	pop2020	pop2021	pop2022	pop2023
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	Los Angeles County, California	709470.0	802542.8	845735.3	880074.7	905024.3	888500.9	25.23445	9996510	9810478	9748524	9732568
2	Cook County, Illinois	251284.0	273716.3	285915.8	298995.5	311439.7	321332.2	27.87609	5263651	5186249	5127866	5142853
3	Harris County, Texas	222698.3	257282.7	291406.6	288550.7	289700.5	282433.8	26.82351	4734060	4736539	4809691	4896507
4	Maricopa County, Arizona	349169.2	449593.8	477538.4	474992.6	476075.7	461212.9	32.08868	4445025	4500059	4559561	4601077
5	San Diego County, California	672439.6	791879.5	862218.2	922096.3	962029.5	939267.3	39.68055	3301110	3272601	3279089	3279736
6	Orange County, California	795519.4	916804.2	1013035.9	1090395.0	1181309.6	1178842.4	48.18525	3186167	3160780	3156252	3149341


```
Call:
lm(formula = hp_change ~ pop_change, data = data)

Residuals:
    Min       1Q   Median       3Q      Max
-84.310  -8.882   2.883  11.083  76.268

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  28.87529    0.33048   87.38  <2e-16 ***
pop_change   1.20463    0.05794   20.79  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 16.88 on 2919 degrees of freedom
Multiple R-squared:  0.129,    Adjusted R-squared:  0.1287
F-statistic: 432.2 on 1 and 2919 DF,  p-value: < 2.2e-16
```

Q3. What is the slope from the simple linear regression?

% change in house price = $0.29 + 1.20 \times \text{\% change in population}$ Explain it in a way laypeople can understand.

For every 1% that a population grew, housing prices increased by an additional 1.2% on average. Population and housing growth move together in a 1 to 1 relationship, higher population higher house price.

Q4. Recreate the chart in the following slide.

It displays the percentage change in house prices on the vertical axis and the percentage change in population on the horizontal axis. Each dot represents an individual county. It includes a trend (regression) line

```
In [5]: library(ggplot2)

ggplot(data, aes(x = pop_change, y = hp_change)) +

  # Reference lines at 0
  geom_hline(yintercept = 0, color = "black", linewidth = 0.5) +
```

```
geom_vline(xintercept = 0, color = "black", linewidth = 0.5) +

# Scatter points
geom_point(color = "blue", alpha = 0.3, size = 1.5) +

# Regression line
geom_smooth(method = "lm", se = FALSE, color = "black", linewidth = 1.2) +

# Equation Label
annotate("text", x = 25, y = 90,
          label = "Price Change = 28.9 + 1.2 * Pop Change",
          fontface = "italic", size = 4.5) +

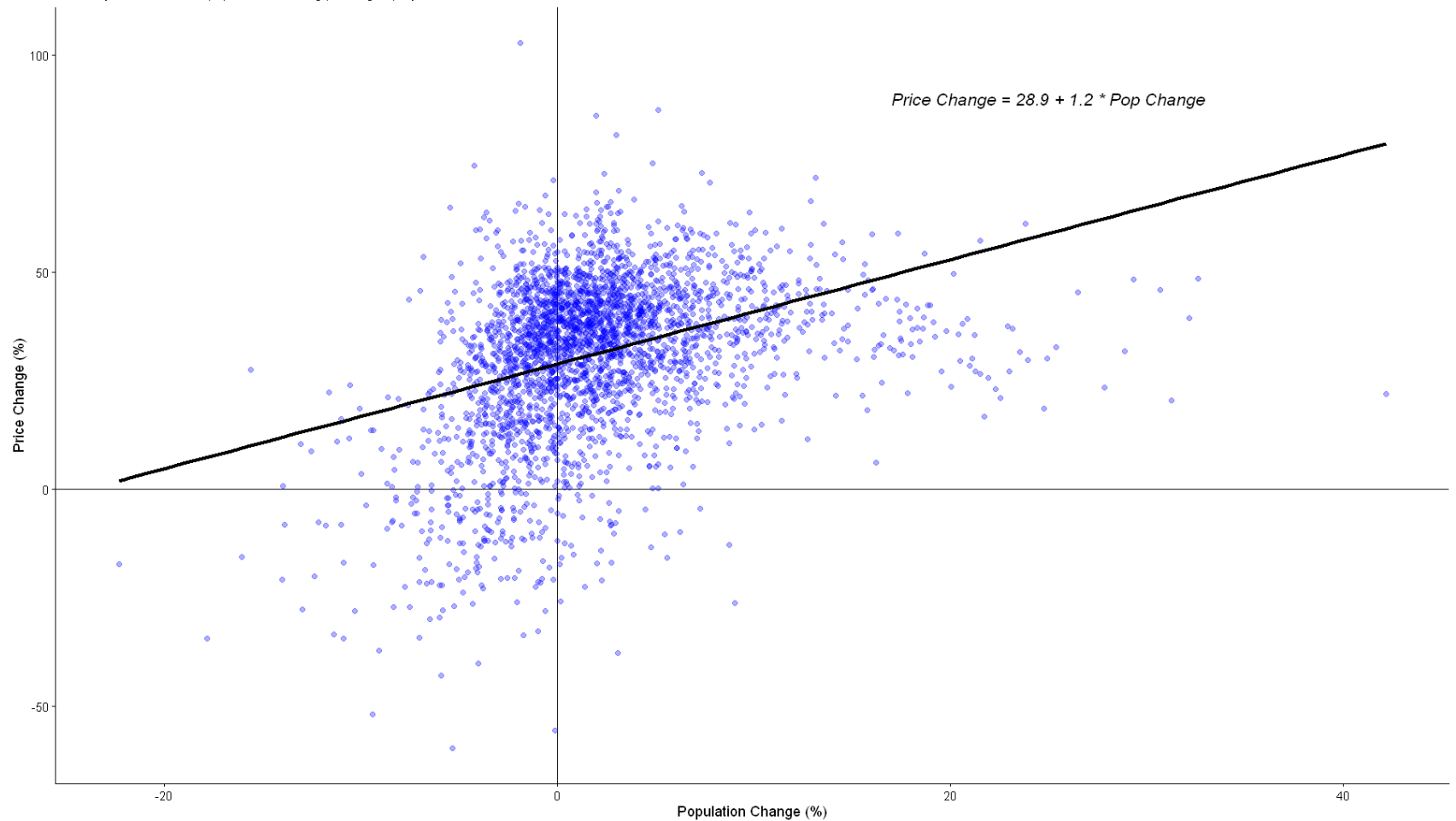
# Axis Labels
labs(
  title = "Housing Price Change vs. Population Change: 2020-2025: County-Level Analysis",
  subtitle = "For every 1% increase in population, housing prices go up by an additional 1.2%",
  x = "Population Change (%)",
  y = "Price Change (%)"
) +

# Theme to match the slide
theme_classic() +
theme(
  plot.title = element_text(face = "bold", size = 13),
  plot.subtitle = element_text(size = 10),
  axis.title = element_text(size = 11)
)
```

```
`geom_smooth()` using formula = 'y ~ x'
```

Housing Price Change vs. Population Change: 2020-2025: County-Level Analysis

For every 1% increase in population, housing prices go up by an additional 1.2%



```
In [6]: # Create counties in Arkansas
ar <- subset(data, grepl('Arkansas', name))
```

```
In [7]: suppressWarnings(suppressMessages({
library(dplyr); library(ggplot2); library(readxl); library(ggrepel); options(warn = -1)

zillow_data = 'https://files.zillowstatic.com/research/public_csvs/zhvi/County_zhvi_uc_sfrcondo_tier_0.33_0.67_sm_sa_
pop_data = 'https://www2.census.gov/programs-surveys/popest/tables/2020-2025/counties/totals/co-est2025-pop.xlsx'

housing = read.csv(zillow_data) %>%
```

```

mutate(State = state.name[match(State, state.abb)],
       name = paste0(RegionName, ', ', State)) %>%
select(name, contains("12.31") & contains("202")) %>%
rename_with(~ paste0("hp", 2020:2025), 2:7) %>% na.omit()

temp_file = tempfile(fileext = ".xlsx")
download.file(pop_data, destfile = temp_file, mode = "wb")
pop = read_xlsx(temp_file, skip = 3)[-1, -2] %>%
  rename_with(~ paste0("pop", 2020:2025), 2:7) %>% rename(name = 1) %>%
  mutate(name = gsub('^\\. ', '', name)) %>% na.omit()
)))

housing$hp_change = ((housing$hp2025 - housing$hp2020) / housing$hp2020) * 100
pop$pop_change = ((pop$pop2025 - pop$pop2020) / pop$pop2020) * 100

data = housing %>% left_join(pop, by = 'name') %>% na.omit()

# Arkansas subset – fix the typo from your original code
ar = subset(data, grepl('Arkansas', name))

# Strip ", Arkansas" for cleaner labels
ar$label = gsub(' County, Arkansas', '', ar$name)

# Fit Arkansas-only regression
model_ar = lm(hp_change ~ pop_change, data = ar)
coefs = round(coef(model_ar), 1)

ggplot(ar, aes(x = pop_change, y = hp_change)) +

  geom_hline(yintercept = 0, color = "black", linewidth = 0.5) +
  geom_vline(xintercept = 0, color = "black", linewidth = 0.5) +

  geom_smooth(method = "lm", se = FALSE, color = "gray60", linewidth = 1.2) +

  geom_point(color = "black", size = 1.5) +

  geom_text_repel(aes(label = label), size = 3, max.overlaps = 100,
                  segment.color = NA, box.padding = 0.2) +

  annotate("text",
          x = min(ar$pop_change, na.rm=TRUE) + 1,
          y = max(ar$hp_change, na.rm=TRUE) - 5,

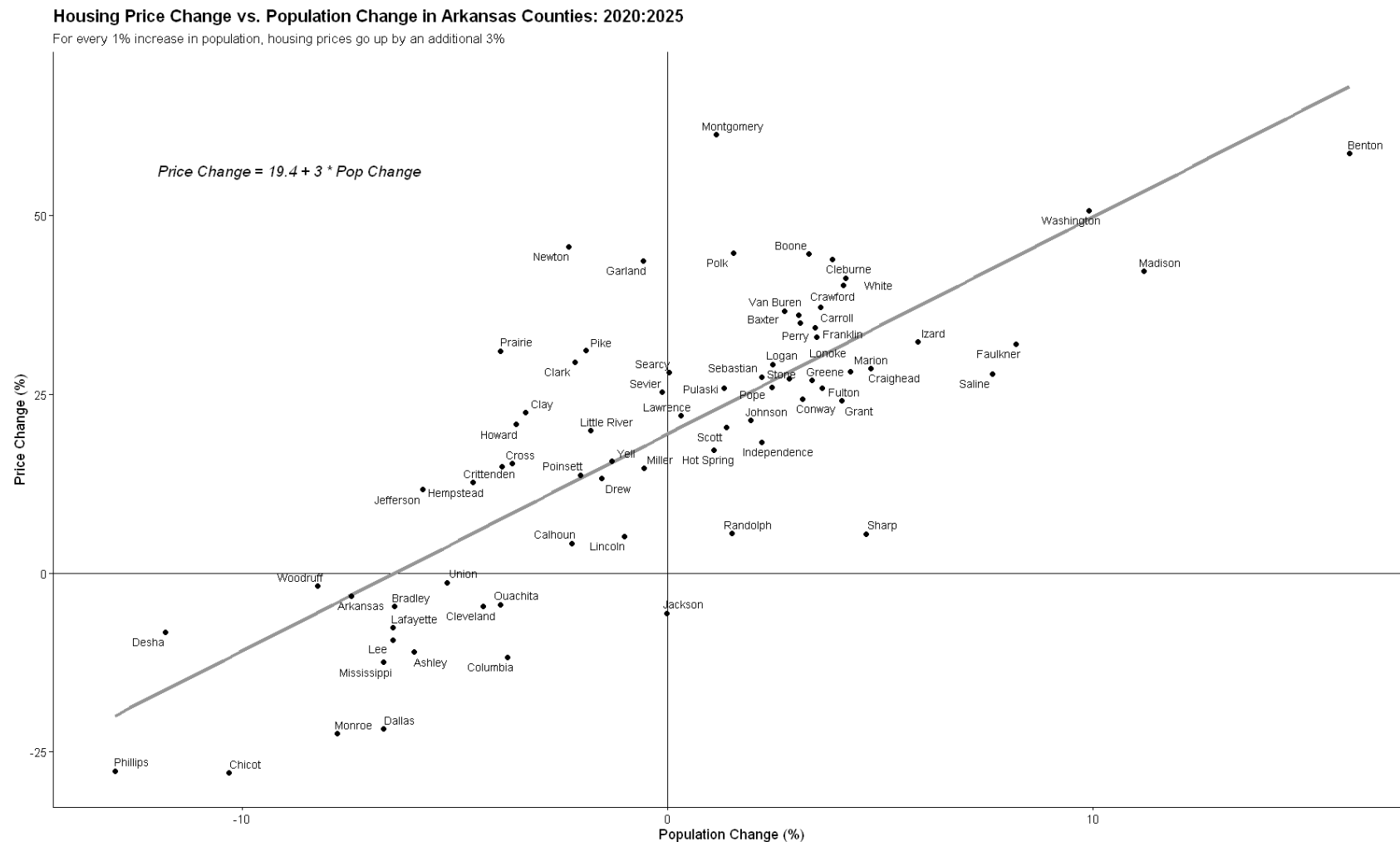
```

```
label = paste0("Price Change = ", coefs[1], " + ", coefs[2], " * Pop Change"),
fontface = "italic", hjust = 0, size = 4) +

labs(
  title = "Housing Price Change vs. Population Change in Arkansas Counties: 2020:2025",
  subtitle = paste0("For every 1% increase in population, housing prices go up by an additional ", coefs[2], "%"),
  x = "Population Change (%)",
  y = "Price Change (%)"
) +

theme_classic() +
theme(
  plot.title = element_text(face = "bold", size = 13),
  plot.subtitle = element_text(size = 10),
  axis.title = element_text(size = 11)
)
```

```
`geom_smooth()` using formula = 'y ~ x'
```



What Does This Tell Us?

The Arkansas slope (**3.0**) is **2.5× steeper** than the national average (1.2).

This means population shifts have a *much larger* impact on home prices in Arkansas than in the typical U.S. county. A likely explanation: Arkansas's housing supply is less elastic than in larger metros — when demand rises (more people moving in), builders cannot quickly add new homes, so prices spike more sharply per unit of population growth.

Counties like **Benton** and **Washington** (the Fayetteville/Rogers metro) stand out as high-growth, high-appreciation outliers, reflecting the tech and logistics boom in Northwest Arkansas. Conversely, many Delta counties lost both residents and home value, consistent with long-run rural depopulation trends.

```
In [8]: # — Extract state name from county name —————
data$state <- gsub('.*', '', data$name)

# — Fit one regression per state —————
state_slopes <- data %>%
  group_by(state) %>%
  filter(n() >= 5) %>%           # need enough counties
  summarise(
    slope = coef(lm(hp_change ~ pop_change))[2],
    n      = n(),
    .groups = "drop"
  ) %>%
  arrange(slope)

# — Add the national (United States) row —————
national_slope <- coef(model)[2]           # already fitted above

state_slopes <- bind_rows(
  state_slopes,
  data.frame(state = "United States", slope = national_slope, n = nrow(data))
)

# — Colour: highlight US and Arkansas —————
state_slopes <- state_slopes %>%
  mutate(
    fill_color = case_when(
      state == "United States" ~ "National",
      state == "Arkansas"      ~ "Arkansas",
      slope > 0                 ~ "Positive",
      TRUE                     ~ "Negative"
    ),
    slope_label = round(slope, 1)
```

```

) %>%
mutate(state = reorder(state, slope))          # sort bars

ggplot(state_slopes, aes(x = state, y = slope, fill = fill_color)) +
  geom_col(width = 0.75) +

  # Reference line at national average
  geom_hline(yintercept = national_slope, linetype = "dashed",
             color = "black", linewidth = 0.7) +

  # Label the national average line
  annotate("text", x = 1, y = national_slope + 0.15,
           label = paste0("National avg = ", round(national_slope, 1)),
           hjust = 0, size = 3.2, fontface = "italic") +

  # Colour scale
  scale_fill_manual(values = c(
    "Positive"      = "#4C8BE0",
    "Negative"      = "#D9534F",
    "Arkansas"      = "#E8A427",
    "National"      = "#2D2D2D"
  ), guide = "none") +

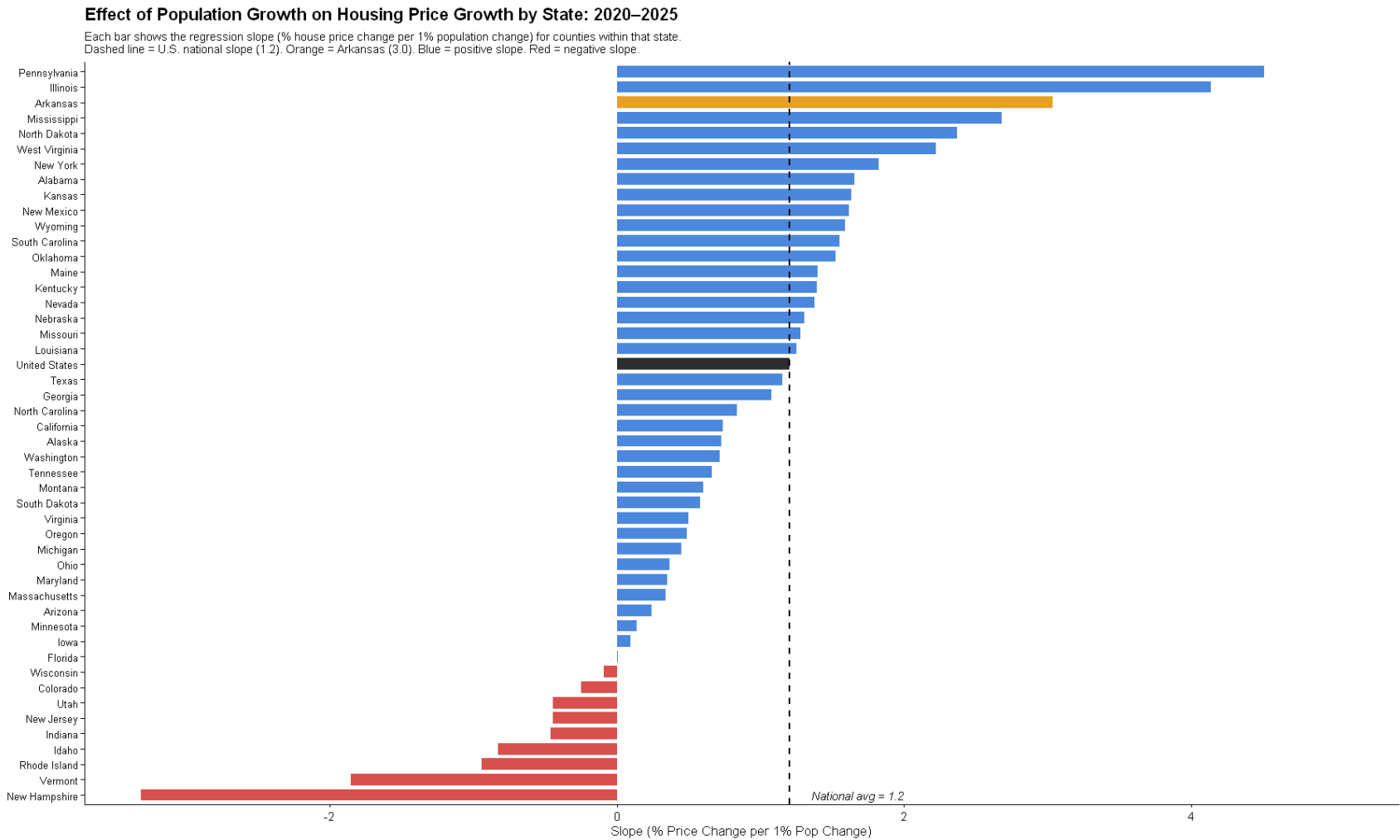
  # Rotate x-axis labels for readability
  scale_y_continuous(expand = expansion(mult = c(0.05, 0.12))) +

  labs(
    title      = "Effect of Population Growth on Housing Price Growth by State: 2020-2025",
    subtitle   = "Each bar shows the regression slope (% house price change per 1% population change) for counties with",
    x = NULL,
    y = "Slope (% Price Change per 1% Pop Change)"
  ) +

  coord_flip() +

  theme_classic() +
  theme(
    plot.title      = element_text(face = "bold", size = 13),
    plot.subtitle   = element_text(size = 9),
    axis.text.y     = element_text(size = 8),
    axis.title.x    = element_text(size = 10)
  )

```

Key Takeaways

Scope	Slope	Interpretation
United States	1.2	For every 1% population growth, home prices rose ~1.2%
Arkansas	3.0	Arkansas home prices are ~2.5× more sensitive to population shifts

What the State Bar Chart Shows

- **Most states have a positive slope** — population growth consistently predicts higher home prices across the country.
- **A few states show negative slopes**, where price dynamics were driven by factors other than population (e.g., remote-work migration reshaping rural areas, oil-price cycles in energy states).
- **Arkansas ranks among the higher-slope states**, reflecting a less elastic housing supply: when people move in, prices respond sharply.
- The **national dashed line** at 1.2 provides a benchmark — states above it are more supply-constrained; those below it have more responsive housing markets.

Limitations

- County-level data averages across very different neighborhoods.
- The model controls for *no other variables* — income growth, interest rates, and zoning laws all matter.
- Connecticut uses planning-region boundaries that don't perfectly match legacy county lines.

Data: Zillow ZHVI & U.S. Census Bureau County Population Estimates, 2020–2025.

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